THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A mixture contains chalk powder, salt and iron filings. The first method to use is: 1

(a) magnet **(b)** filtration **(c)** evaporation **(d)** sieving

2. A student found a dry seed in a drawer. Which test shows that it is living? 1

(a) it is brown **(b)** it is round

(c) it sprouts when given water **(d)** it is light

3. Which of the following is a renewable but exhaustible if misused resource? 1

(a) petroleum **(b)** natural gas **(c)** coal **(d)** forests

4. Which of the following correctly lists planets in order from the Sun? 1

(a) Venus, Mercury, Earth, Mars **(b)** Mercury, Venus, Earth, Mars

(c) Mars, Earth, Venus, Mercury **(d)** Earth, Mars, Venus, Mercury

5. A solution is saturated at 30 °C. On heating to 50 °C it becomes: 1

(a) solid **(b)** gas **(c)** unsaturated **(d)** more saturated

6. A caterpillar changes to a butterfly through a stage called: 1

(a) adult **(b)** egg **(c)** pupa **(d)** seed

7. Which of the following can be reused? 1

(a) ash **(b)** burnt coal **(c)** smoke **(d)** glass bottles

8. A year on a planet is the time it takes to: 1

(a) rise **(b)** complete one revolution around the Sun

(c) cross another planet **(d)** rotate once

9. Why do we heat the salt solution slowly towards the end during evaporation? 1

(a) to dissolve more **(b)** to cool

(c) to freeze **(d)** to avoid spurting

10. A flower produces seeds. Seeds on sowing give plants. This shows: 1

(a) digestion **(b)** excretion **(c)** respiration **(d)** reproduction

11. In 100 L of air, about how much oxygen is there? 1

(a) 78 L **(b)** 1 L **(c)** 50 L **(d)** 21 L

12. Why do we see only one side of the Moon from Earth? 1

(a) Earth blocks it
(b) the other side is dark
(c) it rotates once in the time it orbits the Earth
(d) it never rotates at all

13. After removing iron and adding water to the mixture of chalk and salt, filtering gives chalk as: 1

(a) residue **(b)** solute **(c)** filtrate **(d)** solvent

14. A pebble kept in water for days does not change but a seed swells and sprouts. This shows that: 1

(a) seed is lighter **(b)** seed is living
(c) pebble is heavier **(d)** water is hot

15. Which gas in the air supports burning? 1

(a) argon **(b)** carbon dioxide **(c)** oxygen **(d)** nitrogen

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is evaporation unsuitable to get sand from a sand-water mixture? 2

17. Why is a dry seed considered living although it shows no activity? 2

18. Why is soil called a 'living' resource? 2

19. Why is a day on Earth different from a year on Earth? 2

20. Why can we not separate a mixture of salt and sugar by filtration? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A mixture contains wheat, stones, iron nails and sand. Suggest a plan to separate. 3

22. Explain why fire is not living though it grows, needs air and gives out waste. 3

23. Describe an experiment to show that air is needed for burning. 3

24. Explain, with reasons, why life is possible on Earth but not on the Moon. 3

OR

Explain why sea water is evaporated in shallow pans instead of deep ones. 3

25. Explain why plants do not need food from outside while animals do. 3

OR

Explain how plastics harm the environment. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Plan the separation of a mixture containing sand, sugar, iron filings and water-insoluble sawdust. 5

OR

Plan an experiment to show that plants respond to light and describe expected observations. 5

27. Explain, with the help of an experiment, how to compare the water-holding capacity of sand, clay and loam. 5

OR

Explain how to find the north direction at night using Saptarishi and Pole star, and also use the Moon's phases to estimate the date in the lunar month. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND